Genomics I.

From raw reads to alignments

Orsolya Pipek

Overview

- NGS technology, basic terms and concepts

  PCR, short reads, base calling, paired-end sequencing, quality scores, coverage, etc.

- Data analysis pipeline
  - Alignment
  - Variant calling (SNVs, indels)
  - Interpreting variant files

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Genomics I

Genomics II

next week

Data analysis

A general idea

De-novo assembly

1. Reconstruct the whole genome from the short reads

_diff

is_fa

fairl

rly_d

cult.

_is_f

his_i

this_

fficu

irly_

iffic

De-novo assembly

1. Reconstruct the whole genome from the short reads

```
this_ is_fa  rly_d iffic
 his_i   fairl  _diff  cult.
    _is_f  irly_    fficu
```

(de-novo assembly)

```
this_is_fairly_difficult.
```

Alignment to a reference genome

1. Reconstruct the whole genome from the short reads

even_with_a_template.

th_a_

nplat

_with

even_

a_tem

_tenp

late.

ith_

ven_w

Alignment to a reference genome

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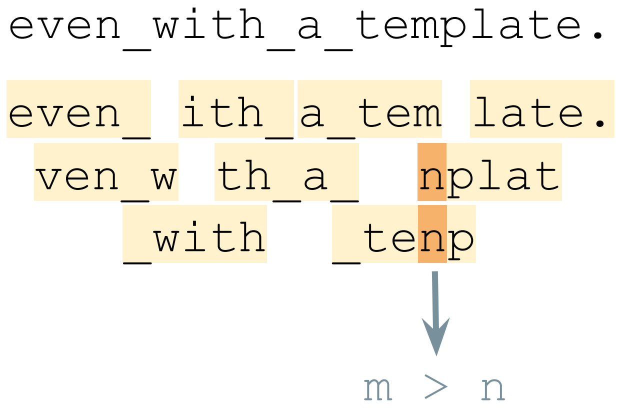
```
even_with_a_template.

even_ ith_a_tem late.
 ven_w th_a_  nplat
    _with  _tenp
```

(alignment to reference genome)

Variant calling

1. Reconstruct the whole genome from the short reads



(alignment to reference genome)

2. Compare the reconstructed genome(s) to a reference genome or to each other and find differences (variants/mutations)

Variant calling

1. Reconstruct the whole genome from the short reads

```
even_with_a_template.

even_  ith_a_tem  late.
 ven_w  th_a_    nplat
    _with    _tenp
```

m > n

(alignment to reference genome)

Not trivial either!

2. Compare the reconstructed genome(s) to a reference genome or to each other and find differences (variants/mutations)

Data analysis

Details

Analysis pipeline: outputs and file formats

NGS	Alignment	Post-processing	Variant calling	Interpretation
Sequences and base qualities of short reads in **random order**.	Sequences and base qualities of short reads with the **genomic position of where they fit on the reference genome**.	**Sorted** (easy to search) alignment files with sequences labelled with **read groups** and sometimes with **duplicates removed**.	**List of somatic and germline variants** with additional information.	List of **annotated, filtered variants**, **figures**, etc.
FASTQ files	SAM/BAM files	BAM files	VCF files	Image files Filtered VCF files

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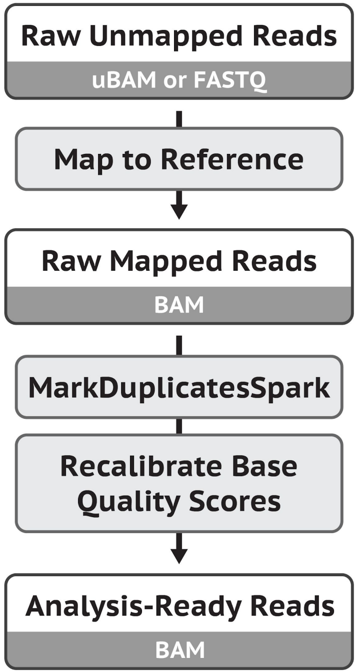
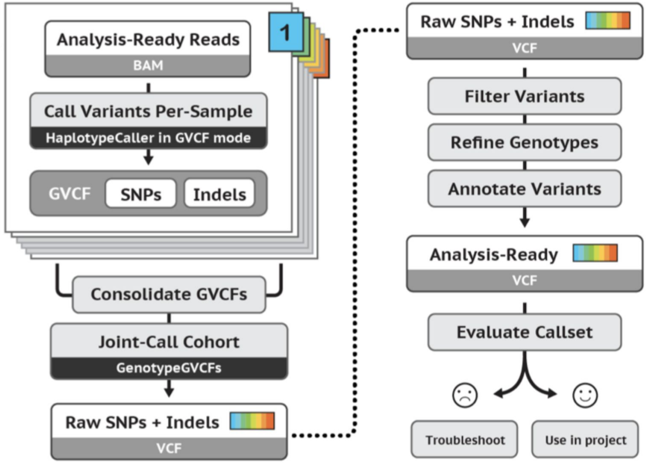
Genomics I ⟵ ⟶ **Genomics II**

GATK best practices

Alignment & post-processing

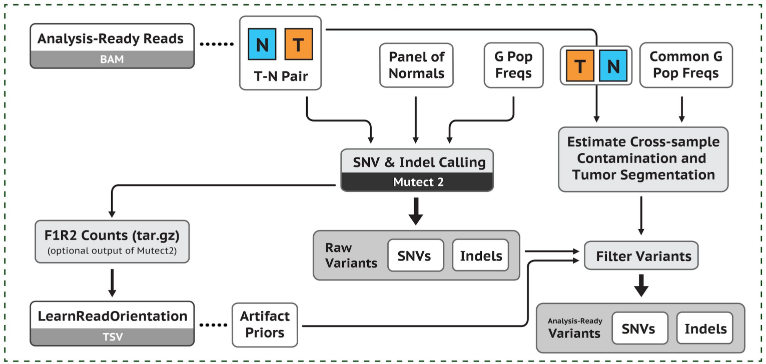




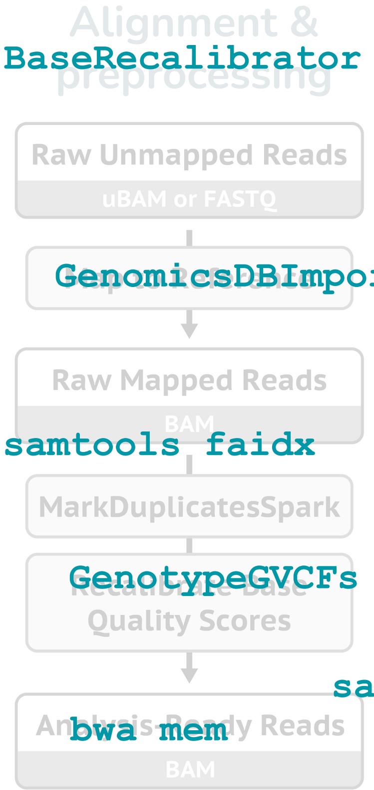
Short germline variants

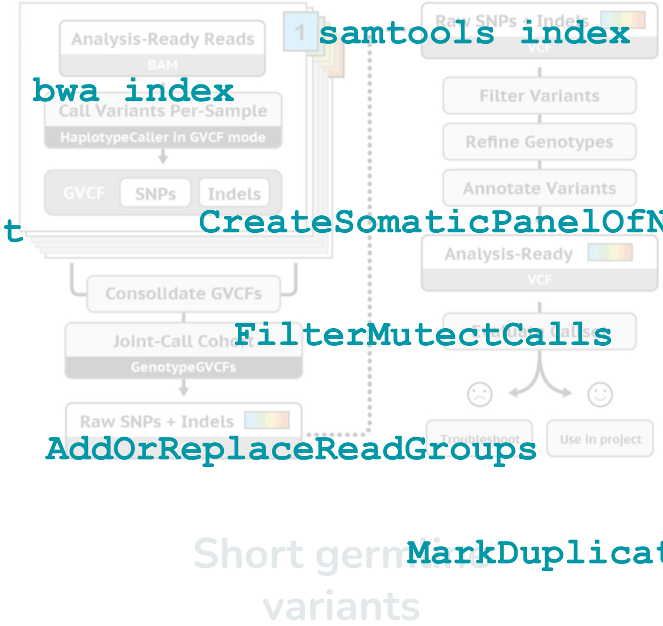
Genome Analysis Toolkit (Broad Institute)

Short somatic variants



GATK best practices





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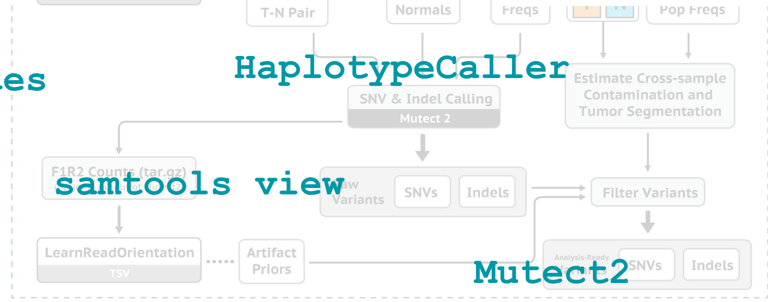

BaseRecalibrator

samtools index

bwa index

GenomicsDBImport

CreateSomaticPanelOfNormals

samtools sort

FilterMutectCalls

samtools faidx

AddOrReplaceReadGroups

GenotypeGVCFs

MarkDuplicates

HaplotypeCaller

samtools mpileup

samtools view

bwa mem

ApplyBQSR

Mutect2

Data analysis

Details

I. **Alignment**

II. Variant calling

Quality control of raw data

Mistakes happen…

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- sequencing is a multi-step process with **many failure points**
- analysis tools **assume reasonable quality input** and can't distinguish real effects from sequencing errors if quality is too low
- poor quality data leads to **false biological conclusions** (e.g. batch effects can look like sample differences, sequencing errors can appear as rare mutations)
- **QC determines appropriate analysis strategies**

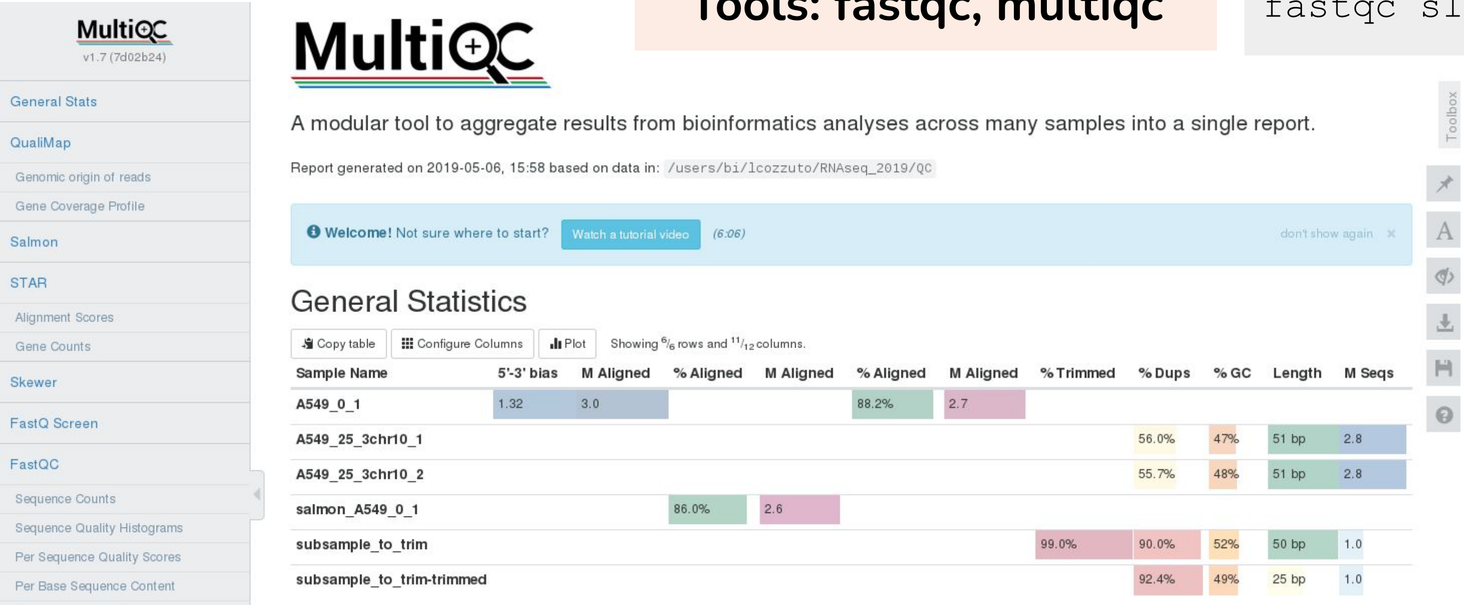
Quality control of raw data

Mistakes happen… all the time, everywhere.

→ before drawing any biological conclusions, it is important to check if the raw data has a decent quality

Tools: fastqc, multiqc

```
fastqc s1.fq.gz
```



Sample Name	5'-3' bias	M Aligned	% Aligned	M Aligned	% Aligned	M Aligned	% Trimmed	% Dups	% GC	Length	M Seqs
A549_0_1	1.32	3.0			88.2%	2.7					
A549_25_3chr10_1								56.0%	47%	51 bp	2.8
A549_25_3chr10_2								55.7%	48%	51 bp	2.8
salmon_A549_0_1			86.0%	2.6							
subsample_to_trim							99.0%	90.0%	52%	50 bp	1.0
subsample_to_trim-trimmed								92.4%	49%	25 bp	1.0

Preparations for alignment

1. Find and download the **appropriate** reference genome (**FASTA format**) (i.e. do not align sequencing data from a chicken to the human reference genome)

   e.g: http://hgdownload.cse.ucsc.edu/goldenPath/hg38/chromosomes/

2. Create an **index file** for the reference genome

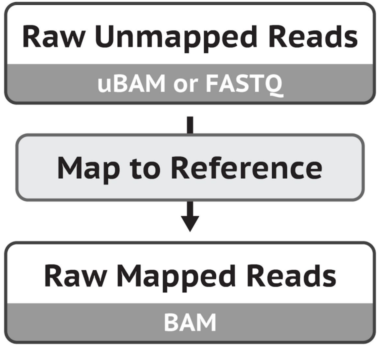
Tools: BWA, samtools

```
samtools faidx refgenome.fa
bwa index refgenome.fa
```

Alignment to a reference genome

Input: short read sequences (and base qualities) in random order (FASTQ files)

Goal: determining the order of the short reads by fitting them to a template (reference genome)



Alignment to a reference genome

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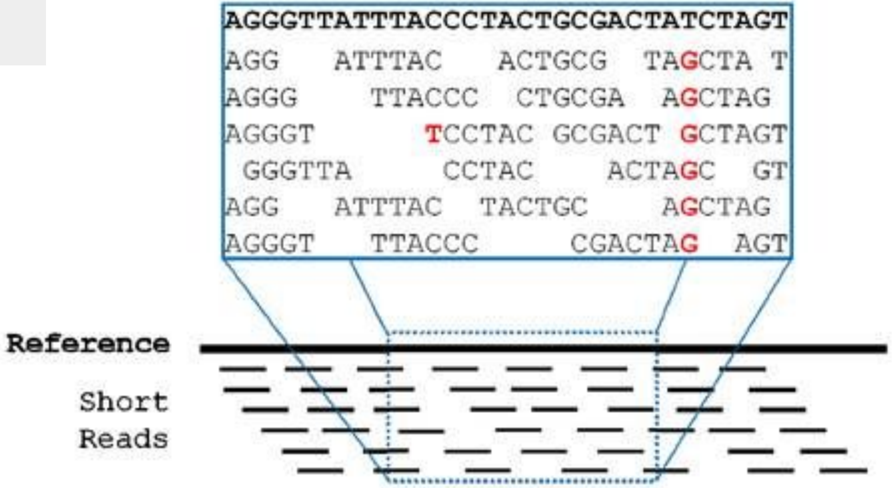
Goal: determining the order of the short reads by fitting them to a template (reference genome)

Tool: BWA

```
bwa mem refgenome.fa s1.fq.gz > s1.sam
```



Alignment to a reference genome

Input: short read sequences (and base qualities) in random order (FASTQ files)

Goal: determining the order of the short reads by fitting them to a template (reference genome)

Tool: BWA

Output: short reads with **position** and **mapping quality** (SAM/BAM files)

Conversion tool: samtools

```
samtools view -bS s1.sam > s1.bam
```

```
1:497:R:-272+13M17D24M     113    1      497    37     37M    15     100338662     0
CGGGTCTGACCTGAGGAGAACTGTGCTCCGCCTTCAG   0;==-==9;>>>>>=>>>>>>>>>>>>=>>>>>>>>>>   XT:A:U NM:i:0 SM:i:37       AM:i:0
X0:i:1 X1:i:0 XM:i:0 XO:i:0 XG:i:0 MD:Z:37
19:20389:F:275+18M2D19M    99     1      17644  0      37M    =      17919  314    TATGACTGCTAATAATACCTACACATGTTAGAACCAT
      >>>>>>>>>>>>>>>>>>>>><<>>>><<>>4::>>:<9   RG:Z:UM0098:1 XT:A:R NM:i:0 SM:i:0 AM:i:0 X0:i:4 X1:i:0 XM:i:0 XO:i:0
XG:i:0 MD:Z:37
19:20389:F:275+18M2D19M    147    1      17919  0      18M2D19M      =      17644  -314
GTAGTACCAACTGTAAGTCCTTATCTTCATACTTTGT   ;44999;499<8<8<<<8<<><<<<><7<;<<<><<   XT:A:R NM:i:2 SM:i:0 AM:i:0 X0:i:4
X1:i:0 XM:i:0 XO:i:1 XG:i:2 MD:Z:18^CA19
```

Alignment to a reference genome

- QNAME: name of the short read present in FASTQ file
- FLAG: a number (“code”) describing the alignment
- RNAME: name of the reference sequence (often contains chromosome)
- POS: mapping position
- MAPQ: mapping quality
- CIGAR string: indicating alignment information
- RNEXT: reference sequence name for the next read
- PNEXT: mapping position for next read
- TLEN: length of read group
- SEQ: short read sequence
- QUAL: short read base qualities
- TAGS: additional optional information

SAM/BAM files

```
1:497:R:-272+13M17D24M     113    1      497    37     37M    15     100338662     0
CGGGTCTGACCTGAGGAGAACTGTGCTCCGCCTTCAG   0;==-==9;>>>>>=>>>>>>>>>>>=>>>>>>>>>>   XT:A:U NM:i:0 SM:i:37       AM:i:0
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          >>>>>>>>>>>>>>>>>>>><<>>><<>>4::>>:<9   RG:Z:UM0098:1 XT:A:R NM:i:0 SM:i:0 AM:i:0 X0:i:4 X1:i:0 XM:i:0 XO:i:0
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19:20389:F:275+18M2D19M    147    1      17919  0      18M2D19M      =      17644  -314
GTAGTACCAACTGTAAGTCCTTATCTTCATACTTTGT   ;44999;499<8<8<<<8<<><<<<><7<;<<<>><<   XT:A:R NM:i:2 SM:i:0 AM:i:0 X0:i:4
X1:i:0 XM:i:0 XO:i:1 XG:i:2 MD:Z:18^CA19
```

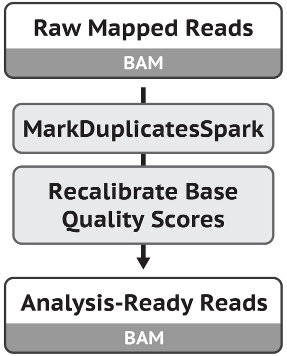
Post-processing alignment files

Input: alignment files (BAM files)

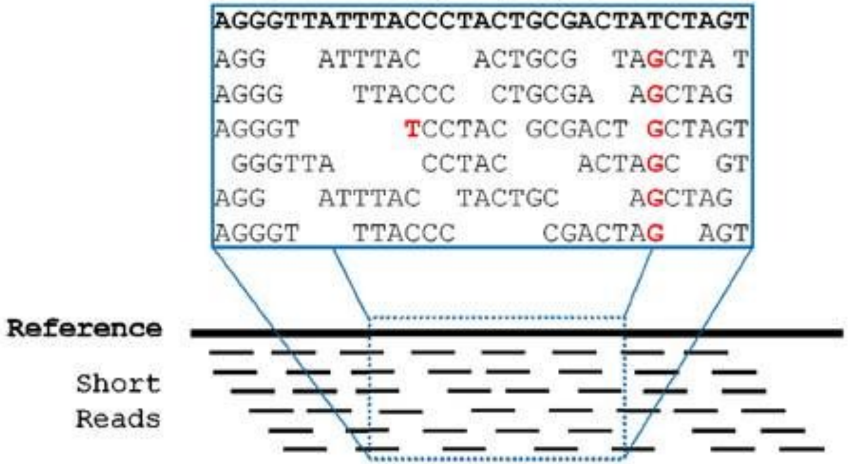
Goals:

- Sorting and indexing BAM files (makes them searchable)
- Removing duplicate reads (generally PCR artifacts, but not always!)
- Adding read groups if necessary (mostly to avoid error messages later)
- Recalibrate base quality scores (more accurate variant calling)

Output: analysis-ready alignment files (BAM files)



Important information in alignment files



- Where were the reads aligned on the reference genome? (position)
- Are they a good match? (mapping quality)
- Do they differ from the reference genome? (reference base, alternate base)
- Can we trust the differing bases? (base quality)
- Are there enough reads to be sure about a difference or is it just chance? (coverage)

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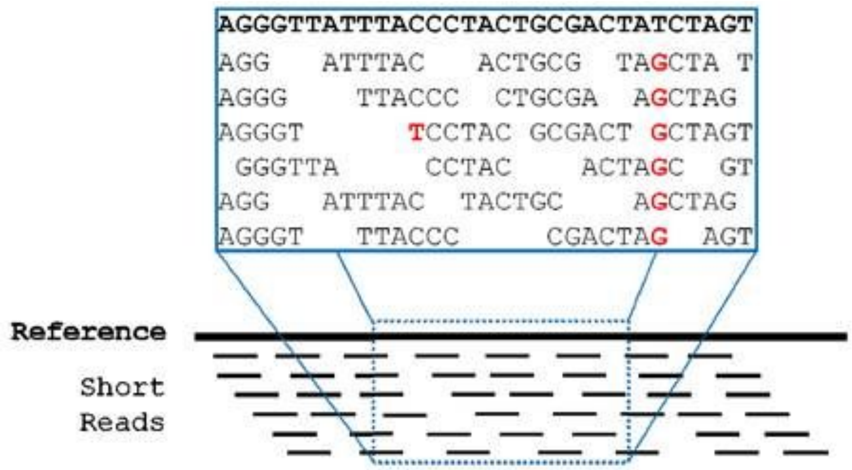

These factors are all considered (besides a bunch of others) during variant detection !

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Data analysis

Details

I. Alignment
II. **Variant calling**

Data analysis

Details

I. Alignment

II. Variant calling

Let's get started!

Step 1

Open tutorial

Step 2

Connect to server

Step 3

Work through tasks

Step 4

Complete and submit Jupyter notebook within 1 week

Get the **tutorial**!

https://pipeko.web.elte.hu/teaching/OMICS/2026_OMICS03_genomics1.html